Hypothesis Testing Framework

Math 742

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Introduction

Motivation and Big Picture

Why Hypothesis Testing Still Matters in Modern Data Science!

In data science, we rarely ask only "What is the parameter value?" Instead, we repeatedly face deployment decisions under uncertainty:

- Should we ship a new model?
- Is a new feature genuinely improving performance?
- Is an observed lift real or just noise?
- Has data drift degraded model quality?

→ These are decision problems, not estimation problems.

Even in settings dominated by prediction and cross-validation, organizations still need:

- Error control
- Reproducibility
- Guardrails against over fitting and false discovery

> Hypothesis testing provides a principled way to make binary decisions using data while explicitly controlling the probability of being wrong.

Scientific Claims vs. Estimation

Estimation answers how much improvement we saw; testing answers whether the improvement is real.

Estimation example:

- "Model B improves AUC by 0.012 on the validation set."

Scientific / operational claim:

- "Model B is better than Model A."

In production systems, that second statement triggers real consequences:

- Engineering time
- Model retraining pipelines
- User-facing changes

- Financial risk

→ Hypothesis testing formalizes the question: Is the observed improvement large enough that we are willing to act on it, given randomness in the data?

Testing as Error-Controlled Decision Making

→ Data science workflows are filled with implicit hypothesis tests, whether we acknowledge them or not.

Examples:

- Comparing two models via cross-validated performance
- Evaluating feature importance or ablation results
- Online A/B tests for recommender systems
- Monitoring performance drift over time

→ Each time we conclude "this change works", we risk:

- Type I error: Deploying a worse model
- Type II error: Failing to deploy a better one

→ Hypothesis testing forces us to:

- Decide which error matters more
- Control the probability of costly false positives
- Make uncertainty explicit rather than anecdotal

This is especially critical in high-throughput settings, where thousands of models, features, or experiments are evaluated.

Scenario: Suppose we have a baseline model M_0 and a new model M_1.

We evaluate both using repeated cross-validation and observe that M_1 has lower average loss. Should we deploy M_1?

```r
set.seed(1999)

n_folds <- 20
true_mean <- 0.50
sigma <- 0.03

# Both models have identical expected performance
loss_M0 <- rnorm(n_folds, mean = true_mean, sd = sigma)
loss_M1 <- rnorm(n_folds, mean = true_mean, sd = sigma)
mean(loss_M1 - loss_M0)
```

```
## [1] -0.00388743
```

```r
mean(loss_M1) < mean(loss_M0)
```

```
## [1] TRUE
```

If we only look at the mean loss for each model, we would conclude that M_1 is better than M_0.

Perform the hypothesis test:

$$H0 : \mathbb{E}[\text{loss } M_1 - \text{loss } M_0] = 0$$

$$H1 : \mathbb{E}[\text{loss } M_1 - \text{loss } M_0] < 0$$

```r
loss_diff <- loss_M1 - loss_M0

t.test(loss_diff, alternative = "less")
```

```
##
##  One Sample t-test
##
## data:  loss_diff
## t = -0.47086, df = 19, p-value = 0.3215
## alternative hypothesis: true mean is less than 0
## 95 percent confidence interval:
##        -Inf 0.01038819
## sample estimates:
##   mean of x
## -0.00388743
```

Here we would not reject the null hypothesis (p-value=0.3215>0.05). We do not have sufficient evidence to conclude the models are different, the difference is just noise.

Summary

- In this simulation, the null hypothesis is true (i.e. both means are 0.5).
- The two models are equally good in the population.
- Any observed difference is purely sampling variability.

→ Hypothesis testing tells us how often we would be fooled if we acted on that variability.

> Even when two models are equally good, one will almost always look better. Hypothesis testing tells us how often we are willing to be wrong when we act on that appearance.

Statistical Hypotheses

Definition 1: A **hypothesis** is a statement about the data-generating distribution (or a subset of the parameter space).

Definition 2: Two complementary hypotheses in a testing problem are called the null hypothesis (H_0) and the alternate hypothesis (H_1)

H_0: Status quo or no difference. For example, adding a new model feature provides no improvement in expected risk.

H_1: There is a difference. For example, adding a new model feature provides an improvement in expected risk.

Mathematically,

$$H_0 : \theta \in \Theta_0 \text{ and } H_1 : \theta \in \Theta_0^c$$

where, Θ_0 is a subset of the parameter space and Θ_0^c is its complement.

Example: Suppose a new manufacturing process is implemented. Let θ denote the average change in the time to make the widget. $H_0 : \theta = 0$ and $H_1 : \theta \neq 0$. This test would be to see if there is a difference.

It is also might make sense to see if the time has decreased. $H_0 : \theta \geq 0$ and $H_1 : \theta < 0$, where $\theta =$New manufacturing time - Old manufacturing time.

Hypotheses that are concerned with the quality of a product are called **acceptance sampling** problems.

Definition 3: A **hypothesis test** is a decision rule that specifies:

- For which sample values the decision is made to not reject H_0.
- For which sample values the decision is made to reject H_0 and accept H_1 as true.

In order to make your decision a **test statistic** is calculated. The test statistic is a function of the sample data (e.g. sample mean).

A **test statistic** is a summary of evidence, such as:

- Difference in cross-validated risk
- Log-likelihood ratio
- Change in loss or accuracy
- Gradient or score statistics

Key idea: The statistic is not the test,the decision rule is.

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Two statisticians can compute the same statistic and make different decisions.

Error Types

When performing a hypothesis test, because we do not know the true value of the parameter we are always in danger of making one of 2 possible errors. These errors are classified as type I and type II.

Definition 4: A **type I error** occurs when we reject H_0 and it's actually true.

The opposite of a type I error is called a type II error.

Definition 5: A **type II error** occurs when we don't reject H_0 and it's actually false.

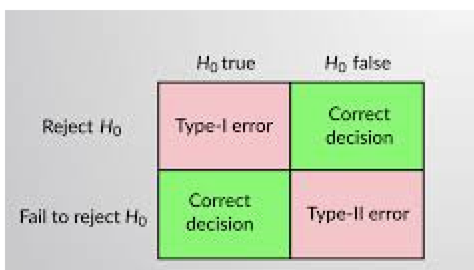

Figure 1: Error Types

Definition 6: The **significance level** (α) of a hypothesis test is the cut-off probability where we deem the result too unlikely to believe the null hypothesis.

The significance level is also the probability of committing a type I error. The following simulation shows that if the null hypothesis is true, and the significance level is 0.05, the null hypothesis will be rejected approximately 5% of the time.

```
# Set parameters
set.seed(12345)           # For reproducibility
n_sim <- 10000          # Number of simulations
n <- 30                 # Sample size
```

```r
alpha <- 0.05          # Significance level

# Store rejection results
rejections <- replicate(n_sim, {
  sample_data <- rnorm(n, mean = 0, sd = 1)  # H0 is true here
  t_test <- t.test(sample_data, mu = 0)       # One-sample t-test
  t_test$p.value < alpha                      # TRUE if we reject H0
})

# Calculate proportion of Type I errors
type1_error_rate <- mean(rejections)
cat("Estimated Type I Error Rate:", type1_error_rate, "\n")
```

```
## Estimated Type I Error Rate: 0.0501
```

```r
cat("Expected (alpha):", alpha, "\n")
```

```
## Expected (alpha): 0.05
```

We prefer not to make a type I error so we could lower the significance level (i.e make burden of proof higher), but then we would reject the null less frequently, so by lowering the significance level we have increased the probability of a type II error.

We handle this dichotomy by determining the consequences of each of these errors and then adjusting α. If a type I error is worse, we might lower α. If a type II error is worse, we might raise α.

Statistics cannot eliminate uncertainty — it redistributes risk.

Summary:

$\mathbb{P}$(Type I error $= \alpha$.

$\mathbb{P}$(Type II error) $= \beta$

Definition 7: The **power** of a statistical test is $1 - \beta$.

Typically we talk about the power of the test instead of the probability of a type II error. We have seen one way to increase the power of the test, increasing α. In a few lectures we will discuss in more detail what affects power. For now, I just summarize the effects on power.

Loss function (implicitly asymmetric)

Implicitly Asymmetric - Type I: deploying bad model = high cost; Type II: missing good one = opportunity cost.

Define the following:

Parameter Space: $\Theta = \Theta_0 \cup \Theta_1$

Hypotheses: $H_0 : \Theta \in \Theta_0,\ H_1 : \Theta \in \Theta_1$

Action Space: $\mathcal{A} = \{a_0, a_1\}$

Where:

$$a_0 : \text{ fail to reject } H_0, a_1 : \text{reject } H_0$$

Let δ be a (possibly randomized) decision rule for the test:

$$\delta : \mathcal{X} \to \mathcal{A}$$

With this notation, we can now define an asymmetric loss function:

$$L : \Theta \times \mathcal{A} \to \mathbb{R}_+$$

$$L(\theta, a_1) = \begin{cases} \lambda_1, & \theta \in \Theta_0, \\ 0, & \theta \in \Theta_1, \end{cases} \qquad L(\theta, a_0) = \begin{cases} 0, & \theta \in \Theta_0, \\ \lambda_2, & \theta \in \Theta_1. \end{cases}$$

where,

- $\lambda_1 > 0$ is the loss of a type I error (false positive)
- $\lambda_2 > 0$ is the loss of a type II error (false negative)

Note: this loss function would be symmetric if $\lambda_1 = \lambda_2$.

The risk function of a test δ would be $R(\theta, \delta) = \mathbb{E}[L(\theta, \delta(X))]$

This means that for,

- $\Theta \in \Theta_0$

$$R(\theta, \delta) = \lambda_1 \mathbb{P}_\theta(\delta(X) = a_1) \text{ (type I error risk)}$$

- $\Theta \in \Theta_1$

$$R(\theta, \delta) = \lambda_2 \mathbb{P}_\theta(\delta(X) = a_0) \text{ (type II error risk)}$$

Hypothesis testing can be viewed as a constrained risk minimization problem: we seek decision rules that minimize Type II error risk, subject to a constraint on the worst-case Type I error risk over the null hypothesis.

$$\text{Minimize } R(\theta, \delta) \text{ for } \Theta \in \Theta_1 \text{ subject to } \sup_{\Theta \in \Theta_0} R(\theta, \delta) \leq \lambda_1 \alpha$$

→ The constraint is what we call the size of the test or the significance level.

Summary and Takeaways

- A Hypothesis test consists of a decision rule with controlled error
- Tests setup controls Type I error
- Power matters, try and maximize it

Takeaways for Data Science PhDs

Hypothesis testing is not about ritualistic p-values. It is about disciplined decision-making in high-dimensional, noisy, data-driven systems.